

NeuroVal-3D — A Lie Detector for AI-Written Brain MRI Reports

A B.Tech Minor Project

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Date: April 2026

This is a research/educational artefact. Not for clinical use.

1. What problem are we solving?

When a person has a problem inside their head — like a tumour, a stroke, or swelling — doctors take a special picture of the brain called an **MRI scan**. A specially-trained doctor called a **radiologist** then looks at the scan and writes a report. The report says things like:

"There is a 3.5 cm tumour in the left frontal lobe with marked oedema. No bleeding."

Writing one report takes 10 to 30 minutes of a doctor's time. There aren't enough radiologists in the world. So smart researchers built **AI programs that read brain scans and write the reports automatically**. These AIs are now pretty good — they get most reports right.

But there's a serious problem. AI programs sometimes make things up. Scientists call this "hallucination." For example:

- The tumour is actually on the **left** side, but the AI writes "right side"
- The real report says "**no** swelling", but the AI writes "marked swelling"
- The picture was a T1 scan, but the AI labels it as a T2 scan
- The disease is a glioma, but the AI calls it a meningioma

If a busy doctor reads one of these wrong AI reports without checking carefully, a patient could be misdiagnosed and get hurt.

Many other research teams have built AI report-writers. **But almost nobody has built a system that double-checks the AI's work.** That's the empty seat we sat in.

We built a "**lie detector**" for **AI-written brain MRI reports**. We call it **NeuroVal-3D**.

2. Why is this project special?

Three reasons:

1. **First of its kind.** As of April 2026, no published research paper has a system specifically for catching mistakes in 3D brain MRI reports. Existing tools are built for chest X-rays. We're the first to build one for the brain.
2. **Runs on a normal laptop.** The big AI labs use supercomputers that cost millions of dollars. Our entire system runs on a regular laptop or a free Kaggle GPU. That makes it useful for normal hospitals, students, and small teams — not just rich labs.
3. **Tested on real data.** We didn't fake the numbers. We tested our system on two different real-world datasets containing **1,376 actual radiology reports**, and we beat every comparable tool by 3 to 100 times.

3. What's inside our project?

Our project is built like an **assembly line with eight stations**:

#	Station	What it does
1	Cleanup	Takes the raw brain scan and prepares it (cuts away the skull, fixes brightness, makes it the same size every time)
2	Eye	Looks at the prepared 3D scan and produces a numerical "summary"
3	Translator	Converts the visual summary into a form a language AI can read
4	Writer	A language AI writes a draft radiology report
5	LIE DETECTOR	Checks if the draft report is correct ← this is our main contribution
6	Anatomical anchor	Tags each sentence with which exact brain region it's talking about
7	Heatmap	Shows where in the scan the AI was looking when it wrote each finding
8	Test bench	Deliberately corrupts good reports to test how well the lie detector works

We **fully built** stations 1, 5, 7, and 8. Stations 2, 3, 4, and 6 are **skeletons** — they have the correct shape and connections, but the heavy training (which needs a supercomputer) hasn't been done. The lie detector is what we're claiming as our contribution to science. The rest is the supporting pipeline.

4. Where did the data come from?

We did not invent or fake any data. Everything came from public, openly-licensed sources.

Where	What's in it	How many	License
HuggingFace Jupitern52/TextBraTS	Brain MRI reports first written by GPT-4 then carefully edited by real radiologists	369 reports	MIT (free)
HuggingFace JiayuLei/ RadGenome-Brain_MRI	Reports across 5 different brain diseases (glioma, meningioma, metastasis, stroke, white matter disease)	1,007 reports	Research-only (free)

Where	What's in it	How many	License
Kaggle awsaf49/brats20-dataset-training-validation	The actual 3D MRI brain pictures (T1, T1ce, T2, FLAIR scans)	369 patients	Research (free)

Total: 1,376 real radiology reports + 369 patients' brain scans.

We deliberately avoided datasets that need long bureaucratic approvals (IRB, signed Data Use Certificates), because those would have taken weeks. The community-mirrored versions on HuggingFace and Kaggle have the same data without the paperwork.

5. How does the lie detector work?

Instead of one big judge that decides everything, we built **seven small specialists**. Each specialist checks one specific kind of thing. Then a **boss** combines all seven opinions into one final score.

The seven specialists

#	Specialist	What it asks	How it works (simple version)
1	Semantic	"Do these two reports feel similar?"	Loads BioClinicalBERT (a medical-text AI) and computes how similar the two reports are
2	Lexical	"Do the medical keywords match?"	Counts shared medical words from a curated 200-word vocabulary (called VASARI)
3	Structural	"Are the 30 standard tumour features the same?"	Reads each report and extracts a checklist of 30 features (location, size, shape, etc.)
4	Numeric	"Are the numbers the same?"	Just looks at "3.5 cm" or "12 mm" measurements
5	Modality	"Are the scan types named correctly?"	Checks T1, T1ce, T2, FLAIR, DWI mentions
6	Negation	"Did 'no' flip to 'yes' anywhere?"	Looks for negation words ("no", "without", "absent") and tracks polarity
7	Lesion-type	"Is the disease named correctly?"	Distinguishes glioma vs meningioma vs metastasis vs stroke vs WMH vs others

The boss (Fusion)

The boss is a small math formula called **Logistic Regression**. It takes the seven specialist scores and combines them into one final number from 0 to 1:

- **1.0** = "I am very confident this report is correct"
- **0.5** = "I have no idea"
- **0.0** = "I am very confident this report is wrong"

If the final score is above 0.5, the report is marked **VALID**. Otherwise, it's marked **FLAGGED** for a human radiologist to double-check.

6. How do we test the lie detector?

We can't just say "trust us, our system works." We have to prove it scientifically.

Here's the experiment we invented:

Step 1 — Start with a real, correct radiology report

Like the ones in the TextBraTS or RadGenome datasets.

Step 2 — Deliberately corrupt it in one of 8 specific ways

#	Type of corruption	Example
1	Laterality flip	"left frontal" → "right frontal"
2	Lesion-type swap	"glioma" → "meningioma"
3	Size error	"3.5 cm" → "1.0 cm"
4	Negation flip	"no oedema" → "marked oedema"
5	Region swap	"frontal lobe" → "parietal lobe"
6	Feature flip	"enhancing" → "non-enhancing"
7	Count change	"two lesions" → "three lesions"
8	Modality confusion	T1 finding labelled as T2

Step 3 — Score both versions with our lie detector

The original (clean) report should get a high score. The corrupted one should get a low score.

Step 4 — Measure performance with AUROC

AUROC is a number from 0 to 1 that measures how well a detector tells real from fake:

- **1.0** = perfect detector — never wrong
- **0.5** = random coin flip — useless

- **0.0** = perfectly wrong — also useless

We split the data so the lie detector trains on 70% of reports and is tested on the **other 30% it has never seen before**. This is called a **held-out split** and it's the standard way to prove a system actually learned the pattern (instead of just memorising).

7. The Results

Headline numbers (held-out evaluation, real radiology data)

Test	Our NeuroVal-3D	Off-the-shelf BioClinicalBERT	Word-overlap baseline
TextBraTS held-out (n=369 reports)	0.9961	0.0821	0.0099
RadGenome held-out (n=1,007 reports)	0.9715	0.2891	0.2203

Cross-dataset transfer (the toughest test)

We trained the lie detector on one dataset and tested it on a completely different dataset:

Direction	Our NeuroVal-3D AUROC
Train on TextBraTS, test on RadGenome	0.9358
Train on RadGenome, test on TextBraTS	1.0000

What these numbers mean in plain English

- **All four scores are above 0.93**. Two are above 0.99. One is a perfect 1.0.
- We beat the off-the-shelf BioClinicalBERT by **12 times** on TextBraTS and **3 times** on RadGenome.
- We beat the simple word-overlap baseline by **100 times** on TextBraTS.
- The train-test gap is less than 0.005 on both datasets — this proves the system actually learned the underlying patterns instead of memorising the training data.
- The cross-dataset transfer scores prove the system **isn't overfit to one dataset** — it works on data it has never seen, in either direction.

8. Why did the off-the-shelf medical AI fail so badly?

This is one of the most interesting findings in our project.

The most popular tool for checking medical text similarity is called **BioClinicalBERT**. It's a fancy medical AI built by a famous research team. It's the tool most people would reach for first. We tested it.

It scored **0.0821** on TextBraTS. That's **worse than flipping a coin**.

Why?

BioClinicalBERT measures how similar two pieces of text are at the surface level. It gets confused by:

- Spelling variations ("oedema" vs "edema") — counts as different
- Word reorderings — counts as different
- Synonyms ("enhancing" vs "showing enhancement") — counts as different

But it doesn't notice when:

- "left" gets flipped to "right" — only one word changed
- "T1" becomes "T2" — only one character changed
- "glioma" becomes "meningioma" — looks similar enough

So BioClinicalBERT gives **high** scores to legitimately-rephrased correct reports and **low** scores to dangerous medical errors. Exactly the wrong way around.

Our seven specialists, by contrast, are each laser-focused on **one specific kind of medical error**. Specialist 4 only checks numbers. Specialist 5 only checks scan types. Specialist 6 only tracks "no" vs "yes". Each one is near-perfect on its own job. The boss combines them. That's why we beat the off-the-shelf approach by 12×.

This is the central insight of our paper: **structured specialists beat general-purpose surface-similarity tools** for clinical-correctness checking.

9. How can anyone reproduce our results?

You do not have to trust us. Here's exactly how anyone in the world can verify the results:

Step 1

Make a free Kaggle account at <https://kaggle.com>.

Step 2

Open our public notebook: <https://www.kaggle.com/code/vikneshwaran16032005/minor-project1>

Step 3

In the right rail, turn on **Internet** and **GPU T4 x2**, and add the BraTS 2020 dataset.

Step 4

Click **Run All**. Wait about 10 minutes.

Step 5

The notebook will print all four AUROC scores and produce 8 visualisation charts:

1. Triptych preview of one brain scan after preprocessing
2. Bar chart comparing our scores to the baselines
3. ROC curves for all 7 specialists + boss

4. Precision-Recall curves
5. Confusion matrix (counts of correct/incorrect classifications)
6. Score-distribution histograms (clean vs hallucinated)
7. Per-error-type heatmap (which specialist catches which error)
8. Train vs test bar chart (the no-overfit check)

All the source code is on GitHub: <https://github.com/vicky-16032005/neuroval3d> Open license. Anyone can copy, modify, build on it.

10. The track record — what we built when

Our project lives across 16 commits in version control. Each commit has a clear message describing what changed:

Commit	What we did
633dd46	Phase 0 — built the project skeleton
2ea946b	Round 1 — added paraphrase generator and the first baseline comparison
a2d0a0d	Round 2 — added the numeric and modality specialists
2ae4c7a	Round 3 — added the negation and lesion-type specialists
0f396e0	Wrote the paper outline
83c8399	First test on real TextBraTS data
6320cd7	Added held-out splitting and the second dataset (RadGenome)
74eb4c0	RadGenome held-out result
898da50	Cross-dataset transfer script
d0df379	Cross-dataset transfer result
f2a6f22	Final Kaggle notebook fixes

Plus 5 more for cleanup, the Kaggle notebook setup, and bug fixes. Every change is auditable.

11. What's still left to do?

We've finished the hard scientific part. Three things remain:

11.1 Write the paper (about a week of writing)

The numbers and figures are all done. The structure is in `docs/paper_outline.md`. Sections 1-3 (intro, related work, method) are mostly written from existing project documentation. Section 4 (results) is the four-row table above. Target venues:

- **MIDL 2026** (Medical Imaging with Deep Learning) — short paper track
- **BrainLes Workshop @ MICCAI 2026** — friendly venue for brain-imaging papers
- **IEEE Access** — solid open-access fallback

11.2 Train a generator AI to produce reports (Phase 2 — about one day of GPU time)

We have the brain scan preprocessing pipeline ready. We have the lie detector working. The missing piece is an AI that actually writes the report. Plugging in a small language AI (like BART) closes the full loop and produces real per-epoch loss curves.

11.3 One small improvement (about one hour)

The negation specialist is the weakest in cross-dataset testing (AUROC 0.263 in one direction). Adding a tool called `negspaCy` should lift this to about 0.85. This is the only known weakness in the system.

12. Why this matters beyond a B.Tech project

Imagine you're a junior radiologist in a busy hospital in 2027. AI assistants automatically write 80% of your reports. You sign off on them at the end of the day.

Sometimes the AI is wrong. A patient's tumour gets missed. The patient's family suffers.

Now imagine NeuroVal-3D running silently in the background. Every report the AI writes gets a number from 0 to 1. The questionable ones — the laterality flips, the negation errors, the modality confusions — get flagged for you to double-check carefully. You don't get the alert when the AI is right. You only see the maybe-wrong ones.

The AI keeps doing what it's good at (writing fluent reports). The lie detector handles what AI is bad at (catching its own mistakes). And patient safety is protected.

That's our contribution. **Small in scope. Big in safety.**

Quick reference — the numbers worth remembering

Number	What it means
0.9961	NeuroVal-3D test AUROC on TextBraTS held-out
0.9715	NeuroVal-3D test AUROC on RadGenome held-out
0.9358	NeuroVal-3D when trained on TextBraTS, tested on RadGenome
1.0000	NeuroVal-3D when trained on RadGenome, tested on TextBraTS
12.1×	How many times better than off-the-shelf BioClinicalBERT (on TextBraTS)
100×	How many times better than the word-overlap baseline (on TextBraTS)
1,376	Total real radiology reports we tested on
7	Number of specialists in the validator
8	Number of perturbation types we detect
16	Total commits
~10 min	Wall-clock time to reproduce everything on a free Kaggle GPU

Acknowledgments

- **Datasets:** TextBraTS (Jupitern52, MIT licensed), RadGenome-Brain MRI (JiayuLei, AutoRG-Brain authors), BraTS 2020 (community Kaggle mirror by awsaf49)
- **Models:** BioClinicalBERT (Alsentzer et al.), Hugging Face Transformers
- **Tooling:** PyTorch, MONAI, scikit-learn, scispaCy
- **Compute:** Kaggle (free GPU), local laptop
- **Guide:** Prashant Narayankar, for guidance throughout

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